STAFF REVIEW · H446 183 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. First: SumTo(1). Last: SumTo(4). The nested calls finish before their callers return.

2. Without a reachable base case, calls continue and the call stack grows until failure.

3. 20, 10, 5.

4. PowerTwo(0)=1, PowerTwo(1)=2, PowerTwo(2)=4 and PowerTwo(3)=8.

I Do — source worked steps

1. Calls — 582 → 58 → 5. Each integer division by 10 discards the last digit.

2. Base case — 5 is a single digit, so CountDigits(5) returns 1.

3. Returns — CountDigits(58)=2; CountDigits(582)=3. The original number has 3 digits.

We Do — guided CHECK

1. Base case
9 is a single digit so returns 1.
Returns
9 returns 1; 90 returns 2; 903 returns 3; 9031 returns 4.

You Do — source mark schemes

1. Calls 67 → 6. Base case returns 1, then the outer call returns 2.

2. Correct: return 1 + CountDigits(n / 10);. The original line does not reduce n, so it never reaches the base case for n >= 10.

3. An example robust implementation: int CountDigits(int n) { long value = Math.Abs((long)n); if (value < 10) return 1; return 1 + CountDigits((int)(value / 10)); } For the most-negative int, the cast is safe here because dividing by 10 reduces the magnitude first.

Exit Ticket — source answer

1. False. Recursion requires a function to call itself; iteration uses a loop.